

- 77** Two flies, which were both heterozygous for a trait, mated and all their eggs were collected. The resulting offspring grew into young adult flies.

The trait is controlled by a single gene with one dominant allele and one recessive allele.

Which of the following statements, taken independently, could be correct for the young adult fly population?

- 1** If the presence of two dominant alleles for this trait stops eggs with this genotype from hatching, the genotypic ratio within the young adult fly population would be 1 : 1.
 - 2** If the presence of two recessive alleles for this trait produces sterile individuals, all of the young adult fly population will have the same phenotype.
 - 3** If the population of young adult flies is small, the phenotypic ratio would be 1 : 2 : 1.
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3